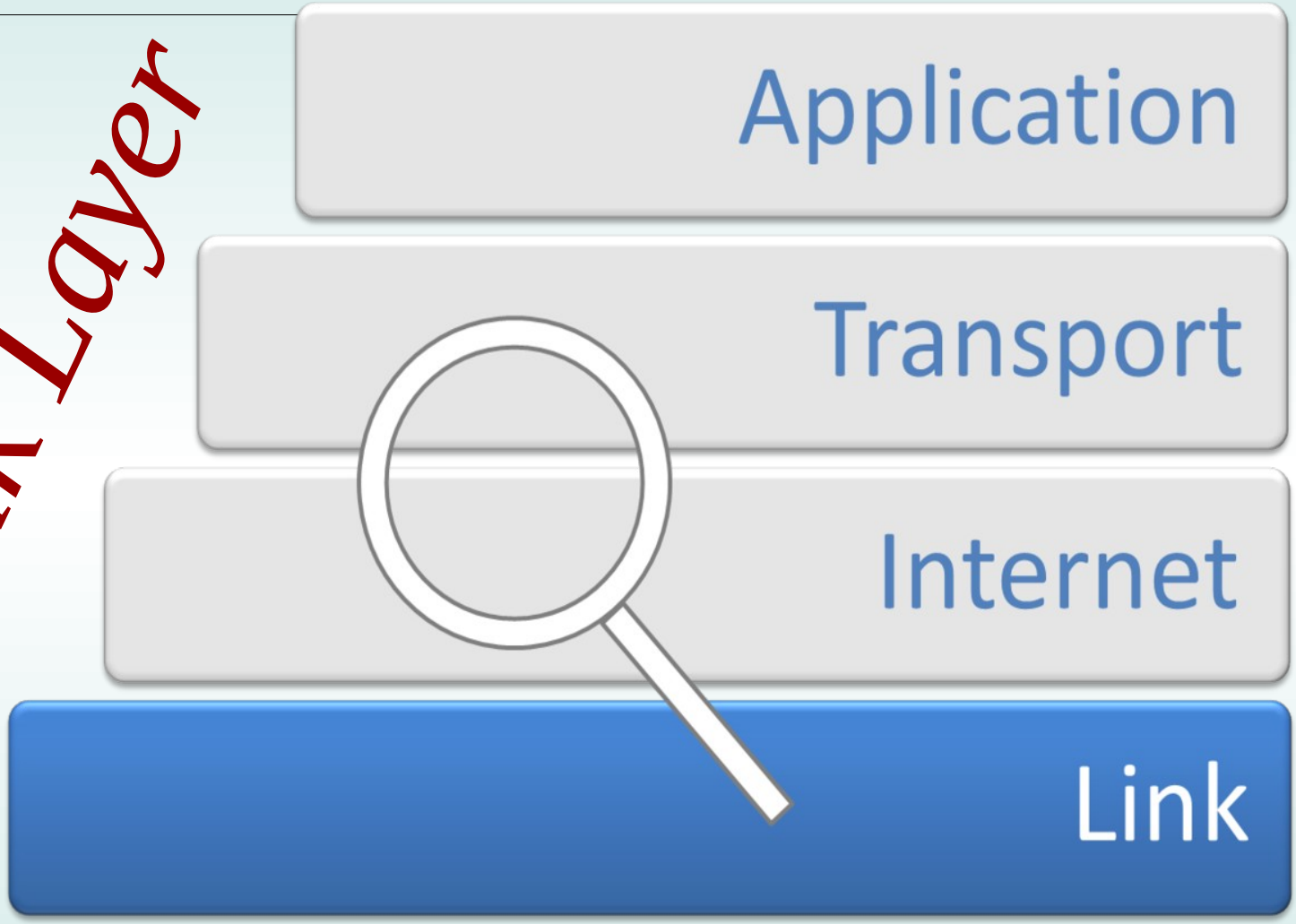
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The Link Layer



CTEC1704 - Networks

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Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs

References

- Kurose & Ross; Computer Networking; Chapter 6
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- https://en.wikipedia.org/wiki/Network_switch
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- https://en.wikipedia.org/wiki/Network_switch
- https://www.tutorialspoint.com/network_security/network_security_data_link_layer.htm
- https://www.cisco.com/c/en/us/td/docs/routers/access/3200/software/wireless/SpanningTree.html

Post lecture discussion topics

- What are the contents of the arp table of the (Linux) computer in front of you?
- Research, name and describe three error detection strategies used by the link layer
- What are the known security issues of the link layer?
- Compare and contrast 10BASE-T with NBASE-T
- What is the current standard?
- Explain the role and functionality of the Spanning Tree Protocol

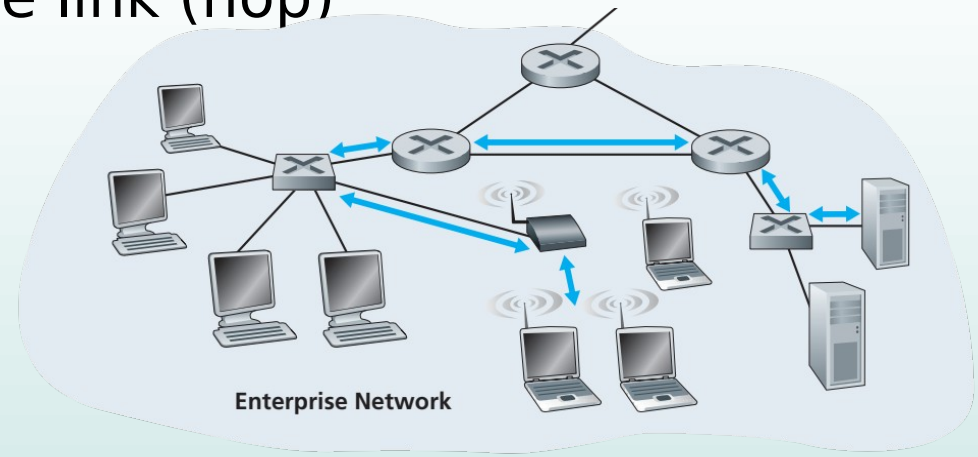
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Lecture Content

- **Principles of Link Layer services**
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs

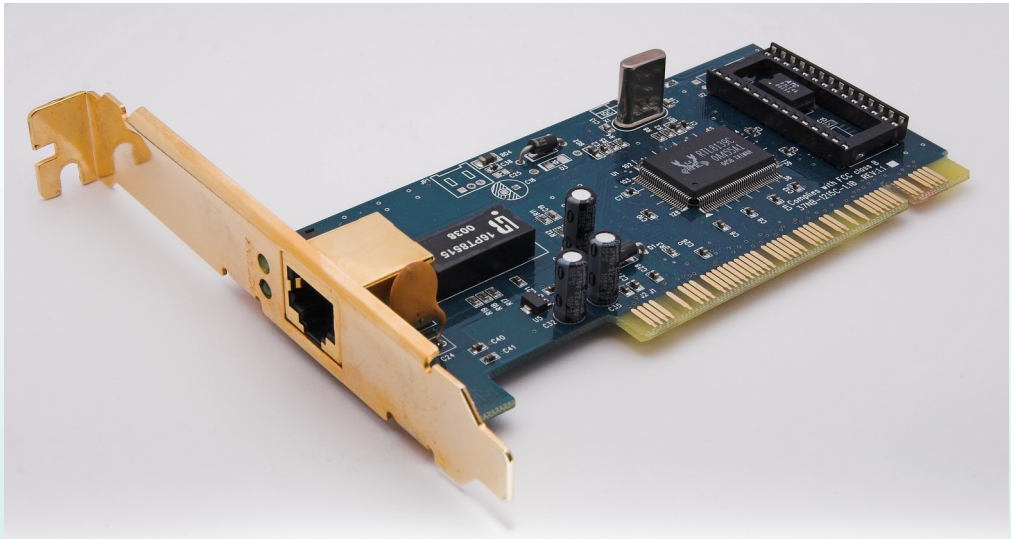
Nodes, Links and hops

- Node -
  - device that runs a link-layer protocol
  - eg hosts, routers, switches, hubs, etc
- Link -
  - comms channel that connects two adjacent nodes
- Link-layer *frame*
  - encapsulates the datagram (network layer) and is transmitted over the link (hop)



Nodes, Links and hops

- Link-layer
  - typically implemented in the Network Adapter or Network Interface Card (NIC)



Link Layer Services

- Frames
  - Link-layer protocols encapsulate datagrams from the network layer before transmission
  - Frame header added to each datagram

Link Layer Services

- Link Access
  - Medium Access Protocols (MAC) specifies how a frame is transmitted onto the link
  - Simple for point to point links
  - More sophisticated when multiple access is required

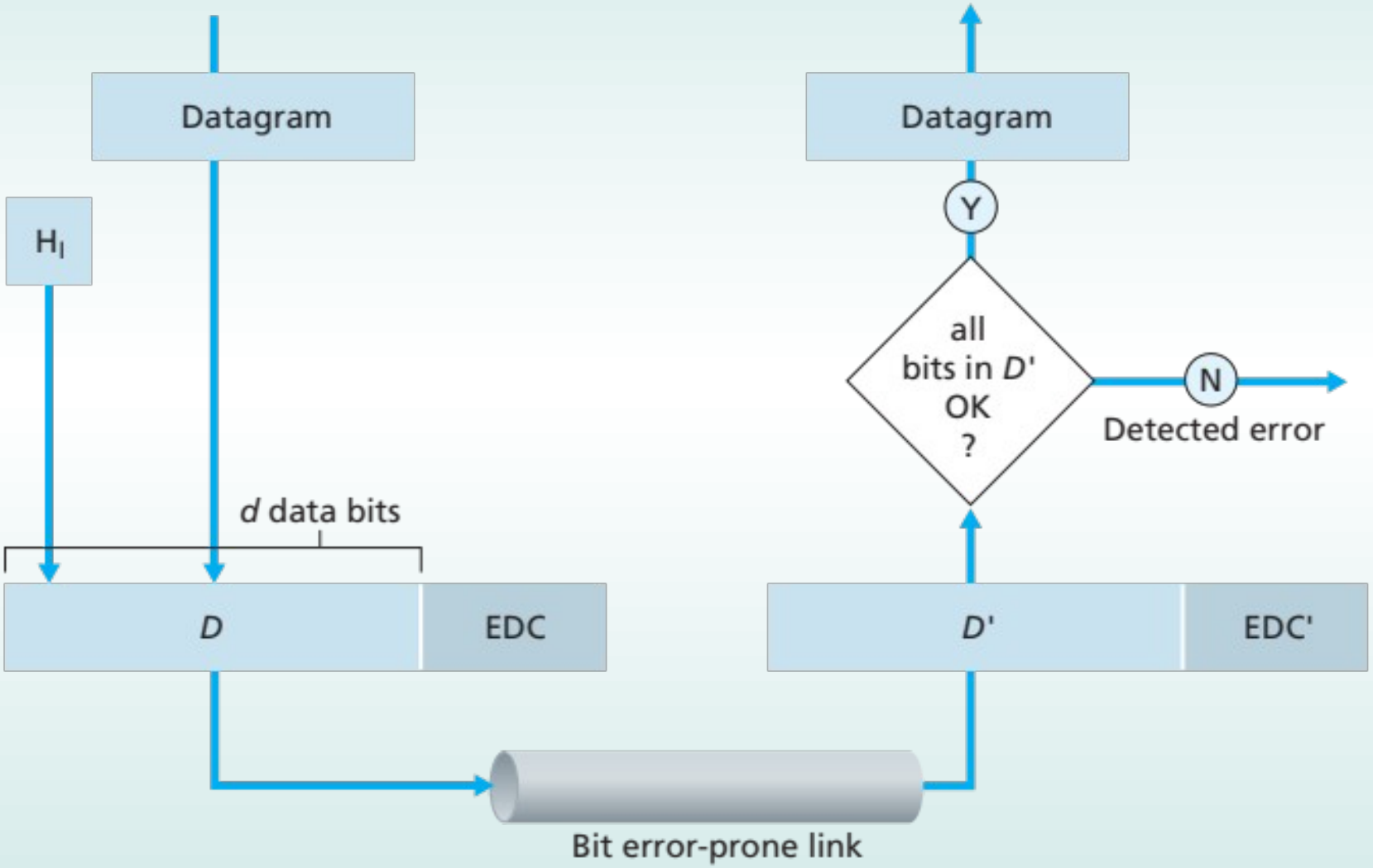
Link Layer Services

- Reliable Delivery
  - Guarantees delivery of each network-layer datagram across the link
  - Typically used when there are high error-rates
    - otherwise, TCP protocol (transport layer) serves a similar function and overheads can be avoided

Link Layer Services

- Error Detection and Correction
  - can detect flipped bits in a frame
    - integrity check
  - typically implemented in hardware and also able to correct errors

Link Layer Services



Error detection

- Types of errors:
  - **Single** bit error
  - In the received frame:
    - only one bit has been flipped (corrupted)
  - **Multiple** bits error
    - more than one bits are flipped
  - **Burst** error
    - more than one consecutive bits are flipped

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Error Detection

- Parity Bits
  - parity bit can detect bit errors
- Cyclic redundancy check
  - CRC
  - similar in concept to the checksum in transport layer
  - more powerful error-detection coding: can detect multiple bit errors

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Lecture Content

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- **Link Layer protocols**
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs

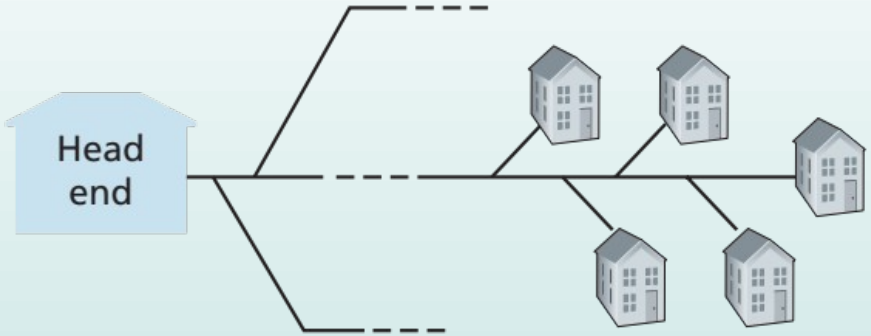
Link Access

- ## Broadcast Link

Multiple sending and receiving nodes connected to a single broadcast channel.
Examples are Ethernet and Wireless LANs

- Problem: How do we make sure the link is used fairly?
- Multiple Access Protocols detect and avoid collision of transmitted frames
- Distinguish Channel Partitioning, Random Access and Taking Turns approaches.

Shared wire
(for example, cable access network)



Shared wireless
(for example, WiFi)



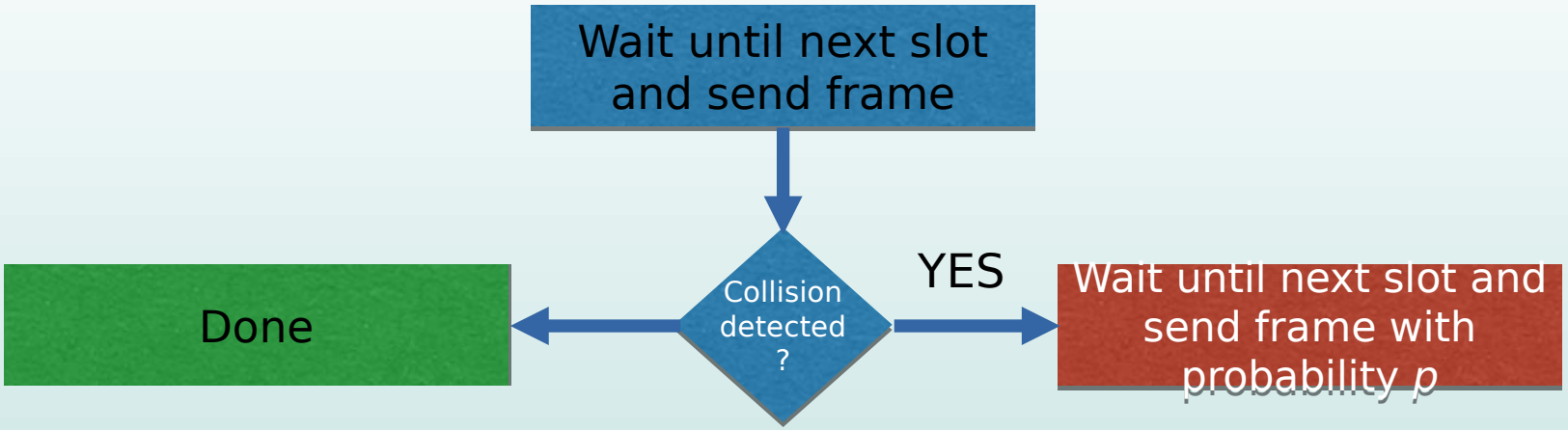
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Link Access

- Time Division Multiplexing
  - divide time into time frames
  - divide each time frame into slots
  - each slot is allocated to a specific node: that can send during its allocated slot only
- Frequency Division Multiplexing
  - divides the channel into different frequencies
  - each node is assigned a frequency
- Code Division Multiple Access
  - each node is given a separate code
  - nodes can submit simultaneously

Random Access

- Slotted ALOHA
  - Frames are equal size
  - Time is divided into equal sized slots (= enough to transmit one frame)
  - Slots are synchronised and nodes transmit only in the beginning of slots
  - When frames collide all nodes detect collision before the slots ends



Carrier Sense Multiple Access (CSMA - CD)

1. Adapter prepares a link-layer frame, and puts the frame in the adapter buffer.

2a. If channel is idle it starts to transmit the frame.

2b. If channel is busy it waits until it senses no signal energy and then starts to transmit

3. During transmission adapter monitors signals to detect collisions.

4a. If adapter transmits the frame without detecting a collision, the frame is sent.

4b. If adapter detects a collision it aborts the transmission and sends a jamming signal.

5. After aborting, the adapter waits a random time and returns to step 2

NB worst case scenario – the packet is never successfully sent!

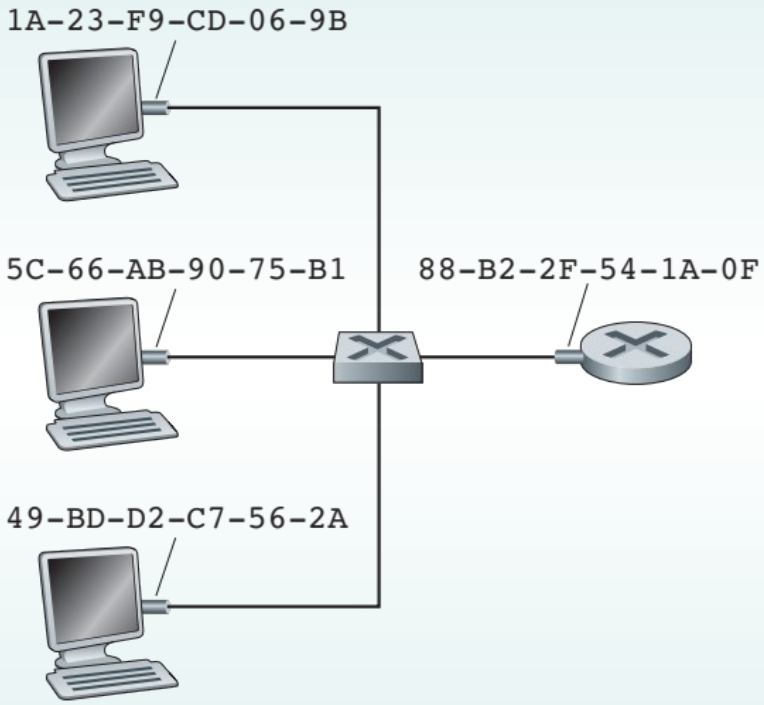
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Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- **Link Layer Addressing**
- Ethernet Protocol
- Switches and Hubs

Link Layer Addresses

- **MAC address** (also called LAN address or physical address)
- **6 bytes** long, typically written as HEX
- first three bytes identify a manufacturer (managed by IEEE)
- last three bytes determined by manufacturer (to produce unique addresses)
- Adapters normally only accept frames for their MAC and discard others.
- Adapters in promiscuous mode reads all frames
- The broadcast FF-FF-FF-FF-FF-FF is read by everyone

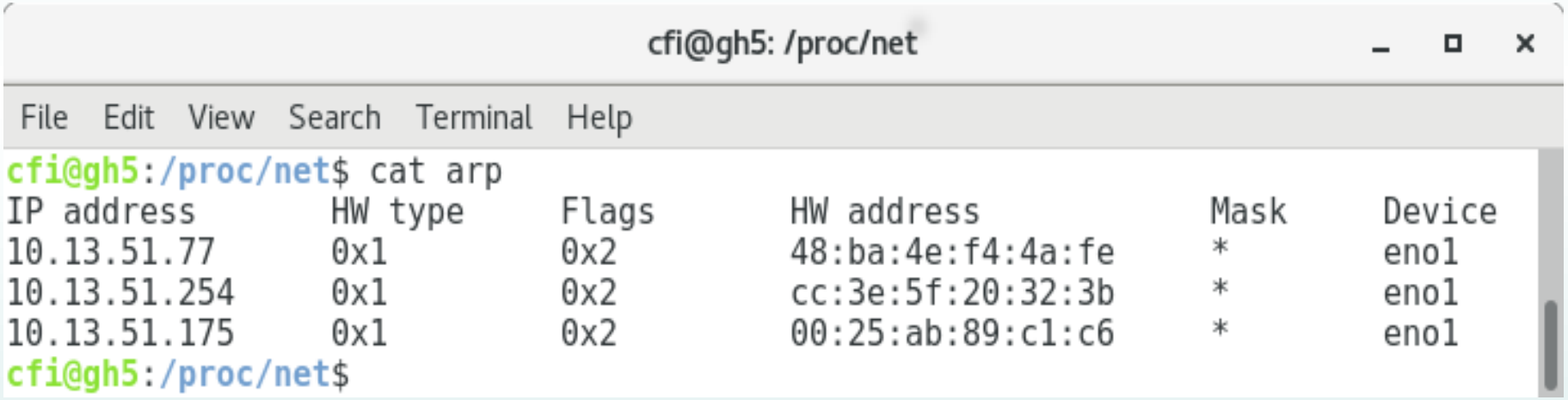


Address Resolution Protocol (ARP)

- ARP resolves an IP address to a MAC address
  - Link Layer: MAC eg 00:1a:a0:b4:4b:c1
  - Network Layer: IP eg 146.227.57.67
- ARP resolves an IP (e.g. 146.227.57.67) to a MAC address (e.g. 00:1a:a0:b4:4b:c1) in the same subnet

ARP Table

- Each host and router maintains an **ARP table** that contains mappings from IP to MAC
- Each entry has an **expiry time**



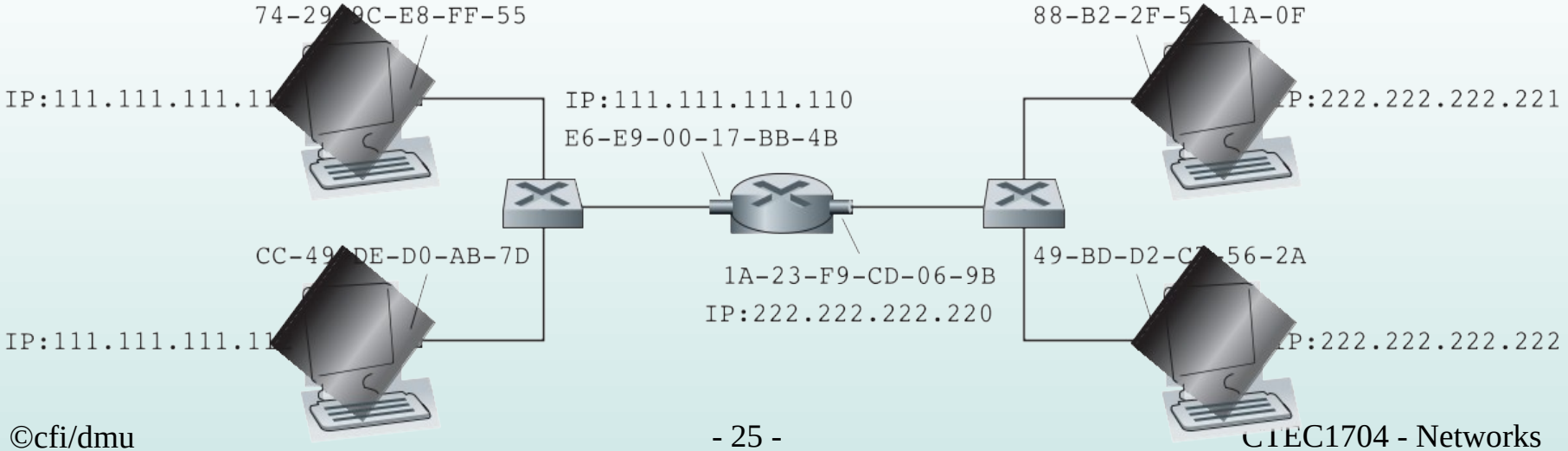
ARP continued

A host sending a datagram to another host/router in the subnet needs to resolve the IP to a MAC address

- If the ARP table contains the MAC address entry it can assemble the frame and transmit
- If the ARP table does not contain the address ARP protocol is used to discover it:
  1. Sender constructs a special **ARP packet**
     - contains header fields that include the sending and receiving IP addresses and the MAC addresses
     - both query and response packets have the same format.
  2. Sender encapsulates the ARP query packet in a frame and transmits to **FF-FF-FF-FF-FF-FF**
     arp 42 who has 192.168.1.1? tell 192.168.1.33
  3. Frame is received by all NICs on the subnet and passed to their ARP module
  4. Each NIC checks if the requested IP address is used by that host
     - If yes, then the NIC replies with the ARP response in a standard frame back to the query IP/MAC
     - If no the ARP query is discarded (or in some cases the IP/MAC is cached in the ARP table)
- Now the sender can update the ARP table and send the datagram

ARP to a remote subnet

- What happens if the datagram needs to be sent off the current subnet?
- Router has an IP/MAC for each of its interfaces
  - (subnets 111.111.111/24 and 222.222.222/24)
- A sender in subnet 1 would use ARP to discover the router 111.111.111.110 with MAC E6-E9-00-17-BB-4B
  - router determines the outgoing interface using a forwarding table
- The outgoing interface (222.222.222.220) passes then the datagram to subnet 2
  - using ARP to discover the destination's MAC before putting the datagram into a frame
  - more ARP in RFC826



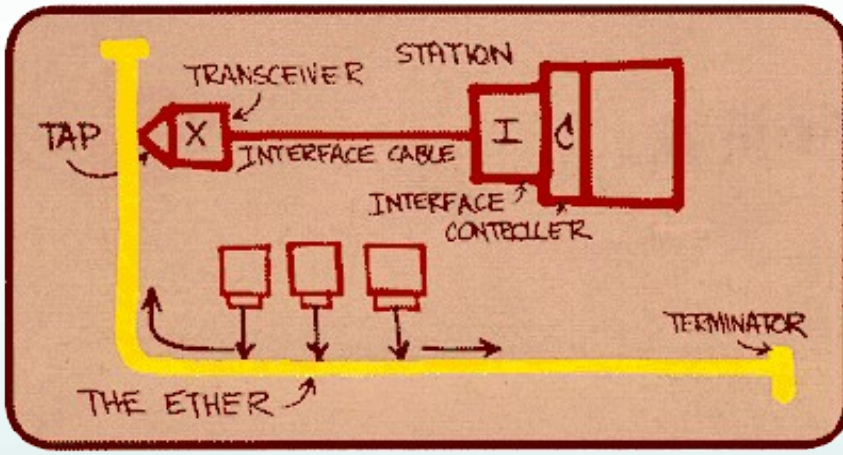
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Lecture Content

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- Link Layer Addressing
- **Ethernet Protocol**
- Switches and Hubs

Ethernet

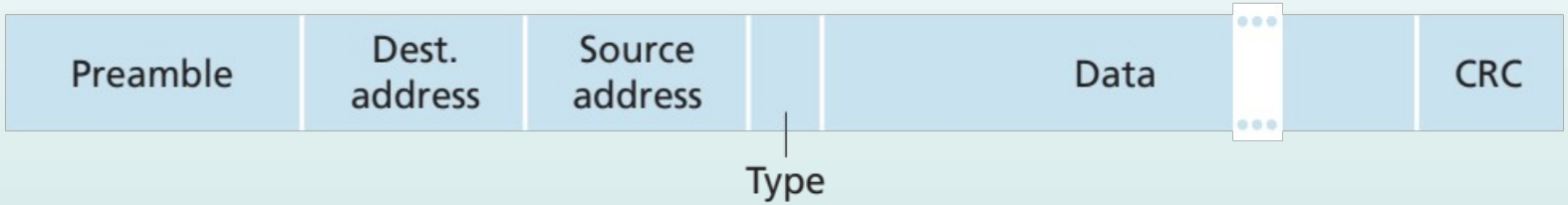
- Dominant wired LAN technology:
  - Cheap @ < £10 for basic NIC
  - kept up with speed race: 10 Mbps – 10 Gbps
  - 400 Gbps soon
  - Terabit Ethernet
    - >1000Gbps



Metcalfe's Ethernet sketch

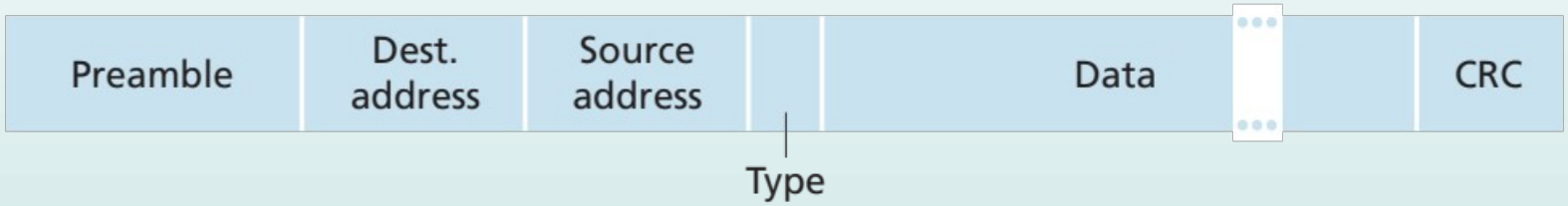
Ethernet Frames

- **Preamble** (8 bytes):
  - AA AA AA AA AA AA AB (used as a wakeup call and for clock synchronisation)
  - The last two bits (...11) are a marker
- **Destination Address** (6 bytes)
  - The MAC address of the destination adaptor
- **Source Address** (6 bytes)
  - The MAC address of the transmitting adaptor



Ethernet Frames

- **Type** (2bytes)
  - value <= 1500
    - size of payload in octets
  - Value >= 1536
    - EtherType – defines which protocol is encapsulated in the payload of the frame
- **Data** (46 - 1,500 bytes)
  - the IP datagram
    - If an IP datagram is larger than 1,500 bytes, it needs to be "fragmented" (network layer)
    - If it is less than 46 bytes it needs to be "stuffed".
- **CRC** (4 bytes)
  - Detect errors in the frame



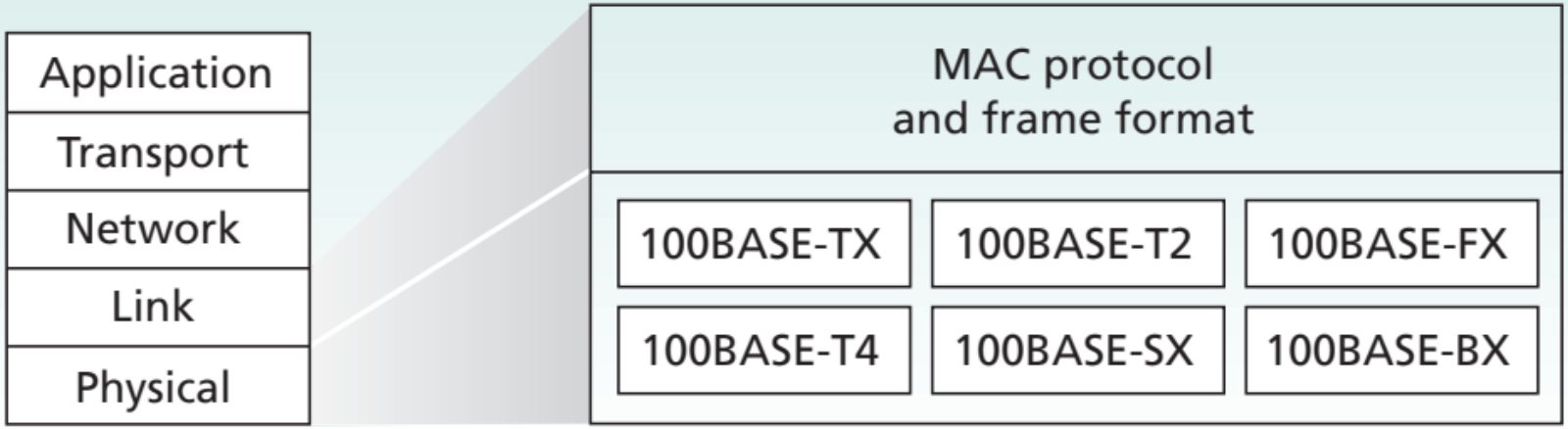
Ethernet Services

- Ethernet provides a **connectionless** service to the network layer
- The ethernet service is "**unreliable**"
  - if a frame fails the CRC check it is simply discarded
  - no acknowledgement is sent

Ethernet Services

- Stream of datagrams (network layer) can have gaps
- These gaps are handled at the transport layer (TCP) or not (UDP)
- Ethernet's MAC protocol
  - unslotted CSMA/CS with binary backoff

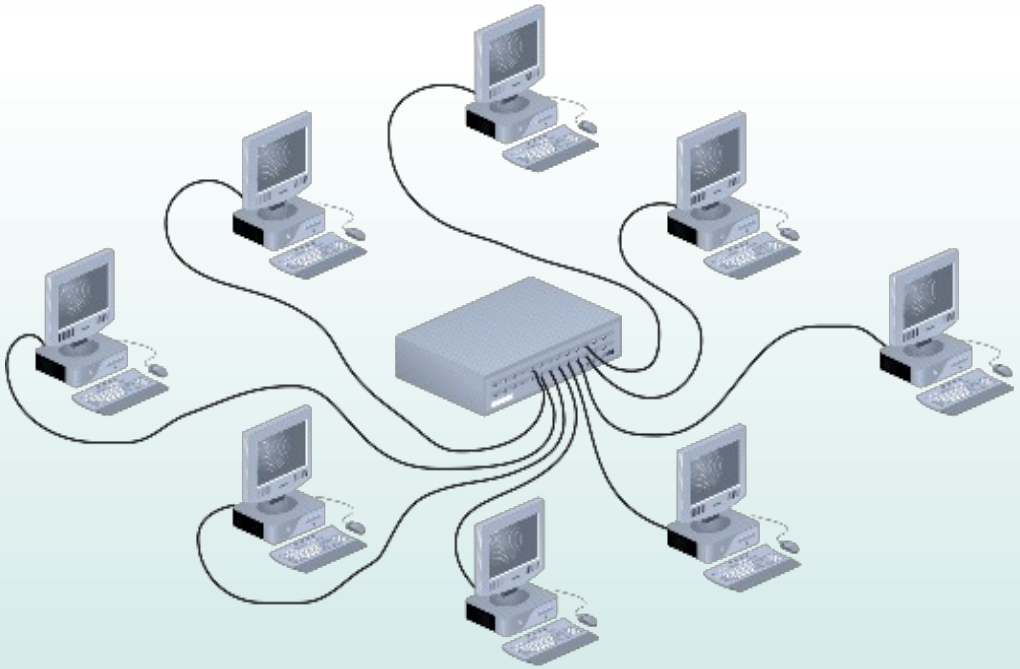
Ethernet Standards: 802.3



- Ethernet is more than Link Layer
  - also specifies the physical layer
- Common MAC protocol and frame format
- Different speeds and physical layer media
- For Gigabit Ethernet with modern full-duplex switches the MAC protocol is becoming obsolete

Ethernet: physical topology

- **Star**
  - prevails today
  - active switch in centre
  - each “spoke” runs a (separate) Ethernet protocol
    - nodes do not collide with each other)



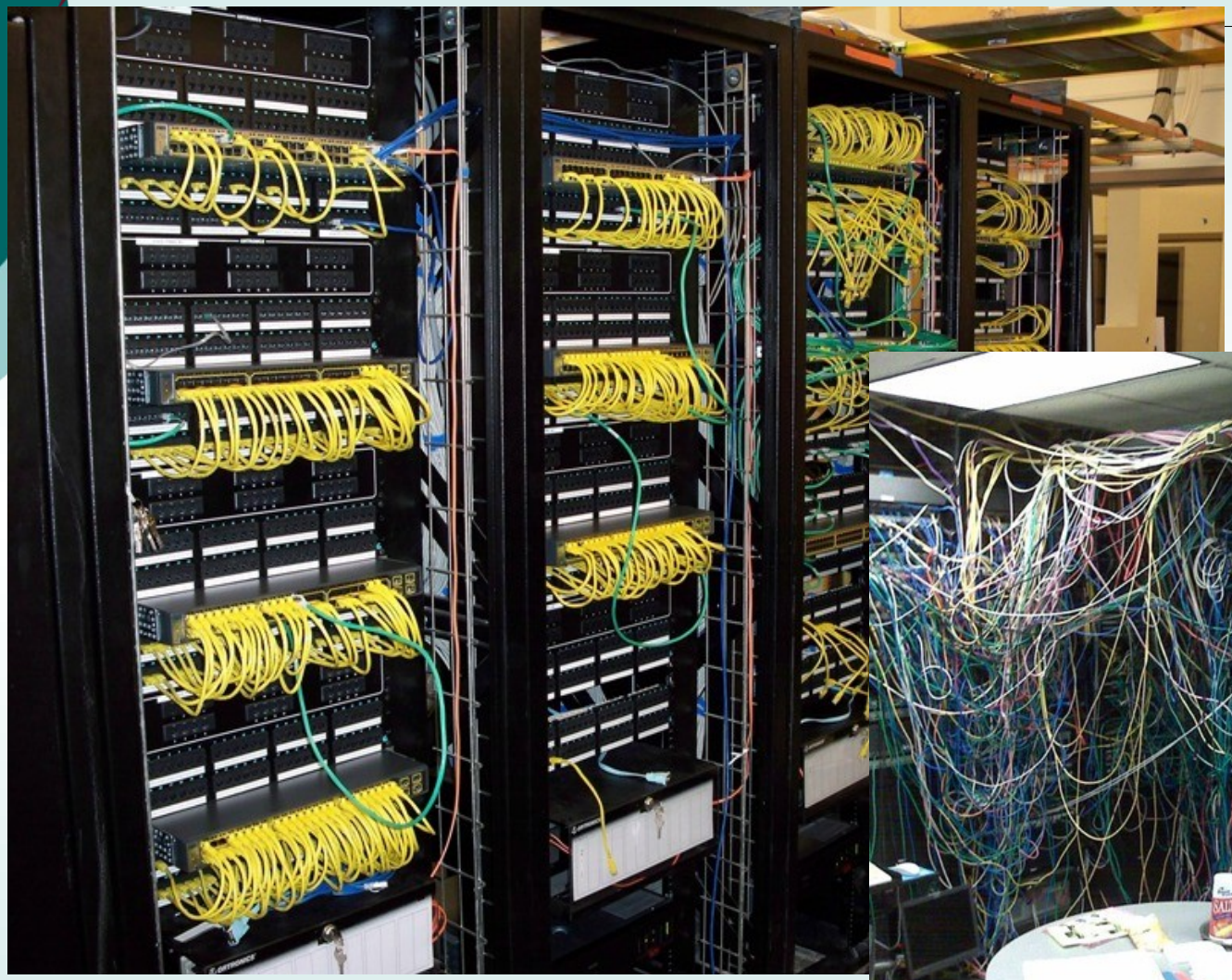
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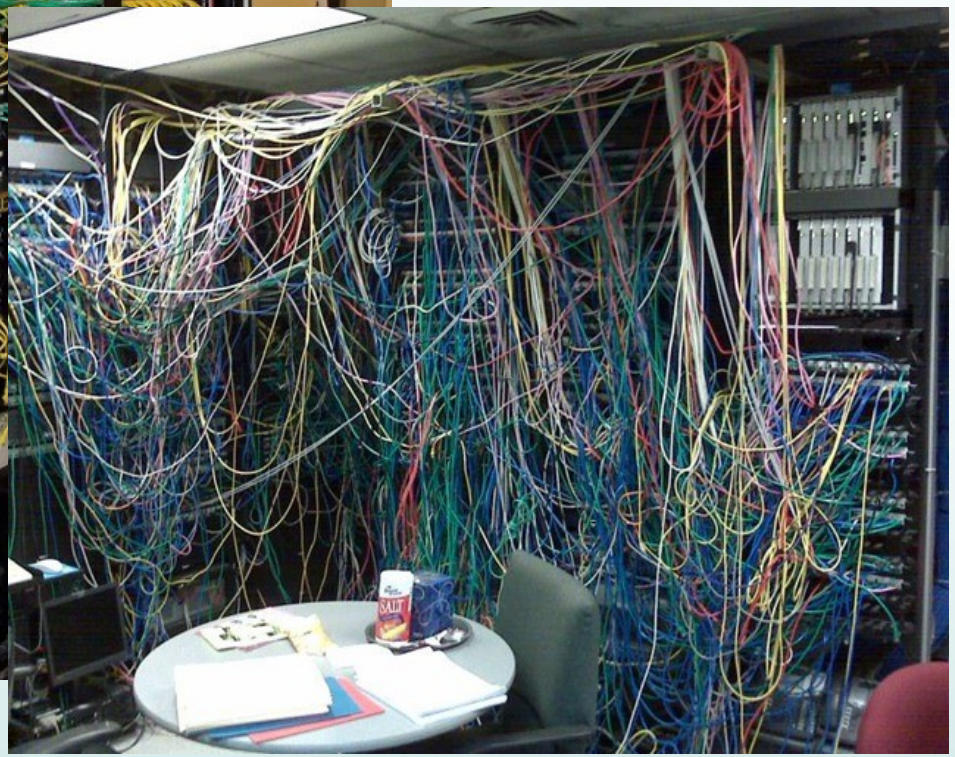
Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- **Switches and Hubs**

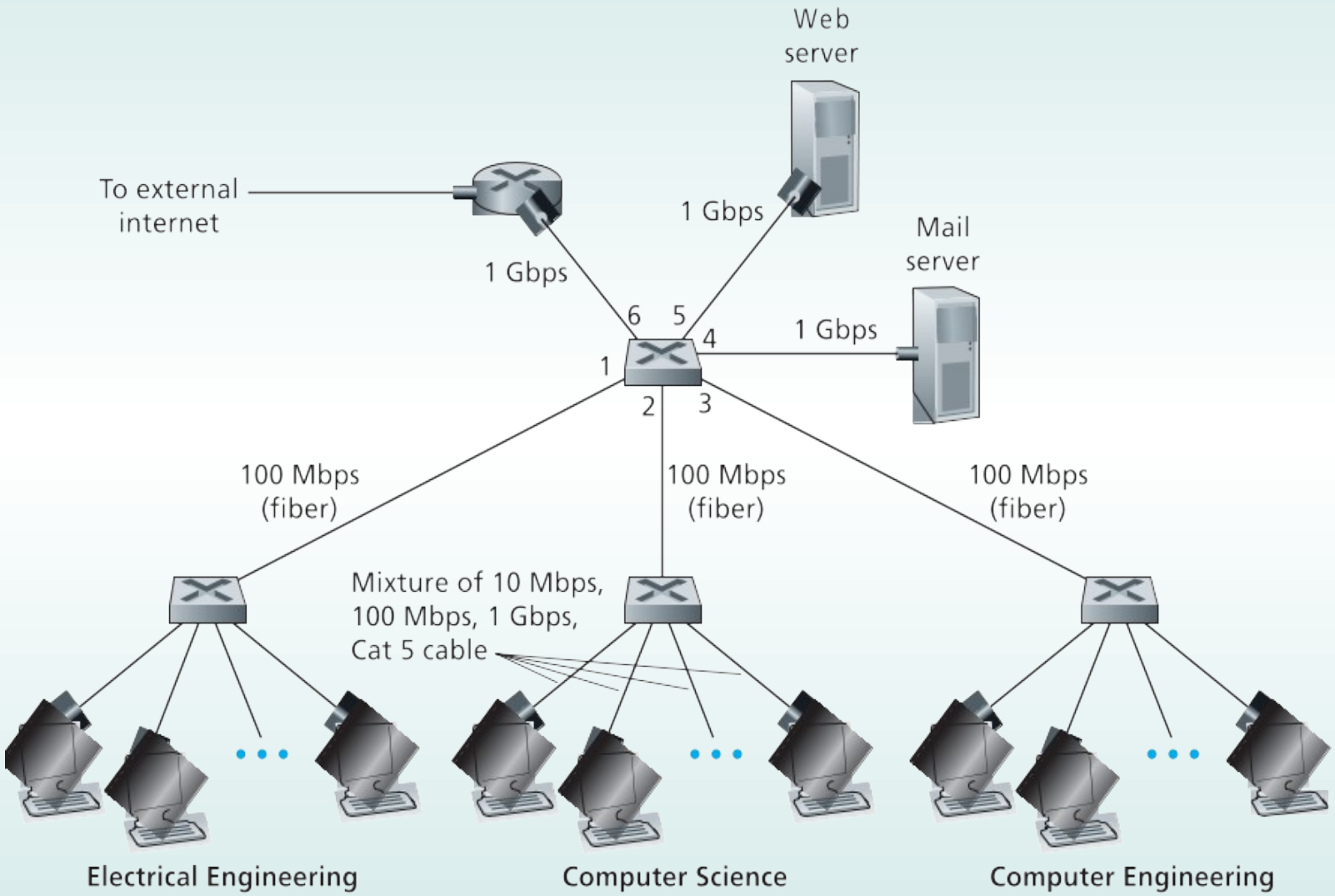
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Switch (cab) room





Link Layer Switches



Link Layer Switches

- Connect networks
- Operate on Link Layer addresses
  - NOT IP addresses
- Remember:
  - Each NIC has a link-layer address
- Switches are transparent
  - they do not have a link layer address

Switches: Filtering and Forwarding

- Filtering determines whether a frame should be dropped
- Forwarding determines the interface to which a frame should be directed and moves the frame to that interface

Switches: Filtering and Forwarding

- This is how the switch decides:
  - If there is no table entry it forwards the frame to all other interfaces
  - If there is an entry and the entry is for the same interface from which the frame originates
    - the frame is filtered
  - If there is an entry and the entry is for another interface than the one the frame originates, then the frame is forwarded to that interface

Address	Interface	Time
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36
....	....	....

Switches: Self-learning

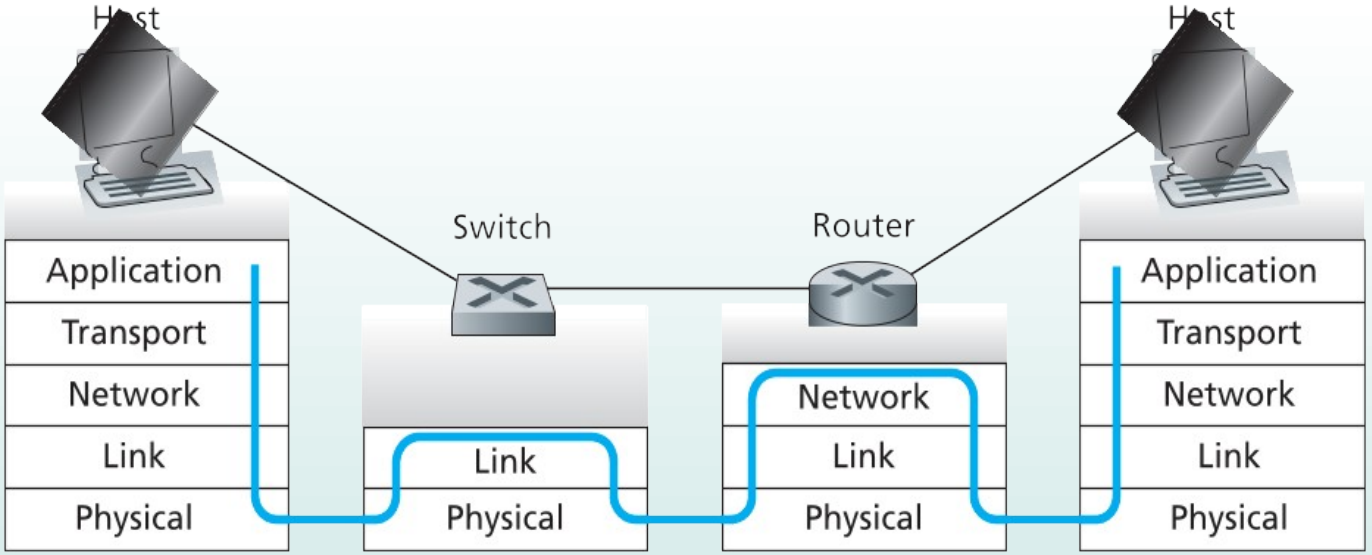
- The switch learns about connected interfaces
  - Initially the switch table is empty
  - For each incoming frame the MAC address from the frame's source address field, the interface on which it arrives and the current time are stored
  - an entry is removed from the switch table after some period of time (Aging Timer)
- No need for manual configuration -> switches are plug and play devices

Link Layer Switches vs Hubs

- A switch learns which MAC is connected to which socket
  - and forwards the frame
- A hub broadcasts a frame to all other ports

Switches vs Routers

- **both are store-and-forward:**
  - routers: network-layer devices (examine network-layer headers)
  - switches: link-layer devices (examine link-layer headers)
- **both have forwarding tables:**
  - routers: compute tables using routing algorithms, IP addresses
  - switches: learn forwarding table using flooding, learning, MAC addresses



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http://global.dlink.com.sg/site_img/Products/DES-3200-Series/DES-3200_main.jpg